

RippleLogic RLS Validation Protocol v1.0

Companion validation protocol for MathGov Core Release 2026.09 v11.4

Status: protocol design, rater-study template, and validation sprint package. This document is not a governing core document, not empirical validation, not legal certification, and not deployment authorization.

Source binding: MathGov is the umbrella framework. RippleLogic is the decision architecture inside MathGov. The public cascade is RG -> RF -> TRC -> CSV -> RLS. The formal shorthand is RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS. RLS ranks only selectable options after Reality Grounding, the Rights Floor, the Tail-Risk Constraint, and Containment and Structural Viability have been satisfied.

1. Executive purpose

This protocol converts the next scientific need for RLS into a concrete study package. The goal is to test whether the 49-cell RLS welfare field is more than detailed. It must become defensible as structured, non-redundant, teachable, repeatable, and useful for better-justified decisions.

- Dimensional non-overlap: test whether the 49 Scope x Dimension cells capture distinct decision-relevant information or collapse into redundant clusters.
- Inter-rater reliability: test whether trained analysts can independently score the same cases with acceptable agreement.
- Decision value: test whether RLS reveals material harms, benefits, near-misses, or trade-patterns that a simpler single metric would hide.
- Revision value: identify which cells, anchors, instructions, or workbook fields need refinement before stronger operational-readiness claims are made.

2. Claim boundary

A successful validation sprint does not prove universal superiority. It can support narrower claims such as: trained raters can apply the RLS scoring manual with measurable agreement on a representative case set; some cells provide distinct information not captured by a scalar score; and identified scoring ambiguities have been logged for Canon or companion refinement.

Until evidence is published or registered, the correct status is: RLS is architecturally specified and audit-ready as a structured residual welfare-ranking layer, but empirical validation of dimensional independence and rater reliability remains in progress.

3. Canonical architecture under test

Layer	Role in validation
RG - Reality Grounding	Claim-authority precondition. Raters must not score claims stronger than the evidence surface permits.
RF - Rights Floor / NCRC	Non-compensatory gate. RLS cannot rescue rights-floor failure.
TRC - Tail-Risk Constraint	Non-compensatory catastrophic-risk gate. RLS cannot average ruin into ordinary benefit.
CSV - Containment and Structural Viability	Selectability gate for containing-system integrity and execution viability.
RLS - RippleLogic Score	Residual welfare-ranking layer over selectable options only.

Core RLS formula: $RLS(a) = \sum_u \sum_d w_u * v_d * m(u,d) * \kappa(u,d) * I_{prop_welfare}(u,d,a)$.

Canonical welfare impact scale: $I(u,d,a)$ in $[-1,+1]$, where 0 means no material change from the declared baseline, not unknown. Unknown active cells must be marked and handled under the missing-data and gate-critical unknown rules.

4. RLS objects to validate

Object	Validation concern	Minimum evidence to collect
Cell score $I(u,d,a)$	Can raters interpret the same cell consistently?	Score, confidence, rationale, evidence basis, uncertainty flag.
Impact magnitude	Can raters build similar magnitude judgments from anchors?	Anchor method, raw indicator, baseline, predicted option value where available.
Union Scope mapping	Do raters map stakeholders to the same scope rows?	Stakeholder-instance map and scope rationale.
Dimension mapping	Do raters distinguish Material, Health, Social, Knowledge, Agency, Meaning, Environment?	Dimension rationale and boundary notes.

Object	Validation concern	Minimum evidence to collect
RLS ranking	Do raters produce comparable option orderings?	RLS or simplified ranking by case, plus decisiveness notes.
Gate relevance flags	Do raters identify RF, TRC, or CSV relevance consistently?	Flag and explanation; not an automated gate verdict.

5. Study ladder

Level	Name	Minimum design	Permitted claim
L0	Internal debug	2-3 cases, 1-2 raters	Only detects obvious rubric or workbook defects.
L1	Validation sprint v0.1	10 synthetic cases, 3 options each, 3-5 raters	Exploratory evidence on clarity, disagreement, and redundancy.
L2	Calibration study v1.0	20-30 cases, 3-5 options each, 5-9 raters	Preliminary IRR and dimensional-overlap evidence.
L3	External shadow-mode study	Real or historical cases, independent raters, preregistered analysis	Domain-specific evidence for operational readiness in shadow mode.
L4	Controlled pilot	Authorized institution, comparator process, outcome follow-up	Limited performance claims relative to a declared comparator.

6. Validation sprint v0.1

- Case set: 10 realistic synthetic cases across AI governance, education, health, environment, public policy, and local infrastructure.
- Options: 3 candidate options per case.
- Raters: minimum 3, preferred 5. Include at least one non-authorial reviewer when possible.
- Scoring surface: 49 RLS cells per option, with score, confidence, rationale, anchor method, evidence basis, and uncertainty flag.
- Blind posture: raters should score independently before discussion.
- Adjudication: after independent scoring, record disagreements and produce an adjudicated reference score only after preserving raw rater scores.

- Analysis: compute IRR, disagreement heatmaps, cell correlations, factor/PCA structure, missingness patterns, and comparison to a simpler scalar or 7-dimension summary.

7. Rater instructions

- Score the change from the declared baseline, not the absolute goodness of the option.
- Do not treat 0 as unknown. Use 0 only for no material baseline-relative change.
- If evidence is insufficient for an active cell, mark UNKNOWN_IMPACT and explain what evidence is missing.
- Do not average away subgroup harms in rights-relevant cells. Flag rights concerns separately from residual welfare scoring.
- Do not let RLS repair an option that fails RF/NCRC, TRC, or CSV.
- Record confidence independently from magnitude. A large impact with weak evidence is not the same as a small impact with strong evidence.
- Use the cell dictionary. When two cells seem similar, explain why the impact belongs in one cell, the other, or both with redundancy handling.
- When unsure, score conservatively, mark the uncertainty, and write the reviewer challenge question.

8. Scoring anchors

Score	Meaning	Use discipline
+1.00	Severe or maximum plausible baseline-relative benefit	Use rarely; requires strong evidence and scope-appropriate denominator.
+0.75	Major durable benefit	Large improvement with clear reach, duration, likelihood, and confidence.
+0.50	Moderate benefit	Meaningful improvement in the cell, not merely cosmetic.
+0.25	Minor benefit	Small but material benefit.
0.00	No material change from baseline	Never use as a substitute for unknown.
-0.25	Minor harm	Small but material harm.
-0.50	Moderate harm	Meaningful degradation in the cell.
-0.75	Major durable harm	Large harmful degradation, even if not gate-failing.
-1.00	Severe harm	Use rarely; often gate-relevant or escalation-worthy.

9. Study hypotheses and failure conditions

ID	Hypothesis	Evidence that supports it	Failure or revision signal
H1	RLS cells preserve non-redundant information.	Cell correlations show related but non-identical patterns; factor structure reveals interpretable clusters without total collapse.	Multiple dimensions correlate above 0.85 across most cases without unique decision contribution.
H2	Trained raters can score cells reliably.	Cell or component ICC meets target or minimum after training; disagreement is explainable and reducible.	Sustained ICC below minimum after rubric training.
H3	RLS improves legibility over scalar scoring.	RLS identifies localized harms, rights-adjacent near-misses, ecological burdens, or scope-level reversals missed by scalar summary.	Scalar model produces same decision and same explanation with no material loss of information.
H4	The manual is teachable.	Raters improve after calibration and can explain cell placement consistently.	Confusion clusters persist in the same cells or dimensions.

10. Statistics plan

- Inter-rater reliability: use ICC(2,1) for single-rater cell scores and ICC(2,k) for averaged panel scores when the same raters score the same case-option-cell units. Use Krippendorff alpha when missingness or mixed measurement levels make ICC fragile.
- Gate and classification reliability: use percent agreement, Cohen or Fleiss kappa, or weighted kappa for binary or ordinal fields such as pass/fail, Prominent versus Guardrail, and uncertainty class.
- Ranking reliability: use Kendall W or Spearman correlation over option rankings within each case.
- Correlation analysis: build a case-option by 49-cell matrix using adjudicated or mean rater scores. Report Pearson and Spearman correlations, clustered heatmaps, and high-correlation pairs.
- Factor/PCA analysis: run only as exploratory unless the sample is large enough. Use it to identify collapse, clusters, cross-loading, and dimensions needing clearer anchors.

- Decision-value analysis: compare final RLS ranking and narrative justification against a simpler scalar score, a 7-dimension-only score, and a 7-scope-only score.

11. Canonical reliability targets

Component	Target	Minimum acceptable	Protocol handling
Magnitude construction mu_k	ICC $\geq$ 0.70	ICC $\geq$ 0.60	Below minimum requires anchor/rubric redesign.
Rights-floor pass/fail	Agreement $\geq$ 0.80	Agreement $\geq$ 0.70	Below minimum requires rights mapping and subgroup guidance refinement.
TRC pass/fail	Agreement $\geq$ 0.80	Agreement $\geq$ 0.70	Below minimum requires scenario and probability guidance refinement.
RLS ranking order	Agreement $\geq$ 0.70	Agreement $\geq$ 0.60	Below minimum requires score, uncertainty, or tie-break review.
PLSS prominence classification	Agreement $\geq$ 0.70	Agreement $\geq$ 0.60	Below minimum requires prominence-signal guidance refinement.

12. Correlation and factor-analysis revision rules

- If two cells correlate above 0.85 across most domains and raters cannot explain distinct causal meanings, mark the pair for overlap review.
- If a cell has persistently low variance across diverse cases, check whether it is genuinely rare, poorly defined, over-masked, or under-evidenced.
- If a dimension shows repeated cross-loading with another dimension, add boundary examples to the scoring manual before considering structural revision.
- If a cell has low reliability but high decision relevance, do not remove it prematurely. Strengthen anchors, examples, evidence requirements, and reviewer checks first.
- If simplified models produce the same winner but lose key localized risk signals, preserve RLS and report the extra legibility as decision-relevant detail.
- If simplified models produce the same winner and same explanation with no material information loss across a larger case set, consider a reduced mode or domain-specific profile.

13. Disagreement taxonomy

Code	Meaning	Likely fix
DGT_SCOPE	Raters disagree on Union Scope mapping.	Improve stakeholder-instance and scope mapping examples.
DGT_DIMENSION	Raters disagree on welfare dimension.	Add boundary examples between dimensions.
DGT_MAGNITUDE	Raters agree direction but not size.	Improve score anchors and indicator scaling.
DGT_SIGN	Raters disagree benefit versus harm.	Clarify baseline, causal pathway, or evidence surface.
DGT_CONFIDENCE	Raters agree magnitude but not confidence.	Improve evidence-quality mapping.
DGT_GATE	Raters disagree on rights, TRC, or CSV relevance.	Strengthen gate trigger guidance.
DGT_UNKNOWN	Raters disagree whether evidence is sufficient.	Clarify unknown and phantom-instance handling.
DGT_DOUBLECOUNT	Raters place same effect in multiple cells inconsistently.	Add redundancy-handling examples.

14. Case packet standard

- Decision boundary and baseline.
- Option set with 2-5 options.
- Stakeholder-instance map and active Union Scopes.
- Evidence packet, including what is known, unknown, contested, and assumed.
- Rights Floor, TRC, and CSV pre-screen notes.
- RLS scoring sheet for every option.
- Rater confidence and rationale fields.
- Adjudication sheet that preserves raw scores before consensus.
- Post-analysis revision notes.

15. Synthetic case roster

Case	Title	Purpose
C01	AI tutor in public schools	Compare mandatory deployment, opt-in teacher-supervised deployment, and no deployment for a public-school AI tutor.

Case	Title	Purpose
C02	Flood relocation policy	Compare voluntary relocation support, mandatory relocation, and infrastructure-only flood protection for a high-risk community.
C03	Agricultural pesticide decision	Compare current pesticide use, restricted integrated pest management, and rapid ban with transition subsidy.
C04	Hospital triage support system	Compare clinician-only triage, advisory AI triage, and automated triage for non-emergency scheduling.
C05	Social-media moderation rule	Compare minimal moderation, rights-protecting graduated moderation, and aggressive automated removal.
C06	Welfare fraud detection model	Compare manual review, AI risk flagging with human appeal, and automated benefit suspension.
C07	Urban traffic redesign	Compare car-priority road widening, bus/bike corridor redesign, and congestion pricing with equity rebates.
C08	Renewable microgrid project	Compare no project, community-owned solar microgrid, and vendor-owned microgrid with long lock-in contract.
C09	School phone policy	Compare unrestricted use, blanket ban, and structured phone zones with exceptions and student voice.
C10	Local data center proposal	Compare approval as proposed, approval with water/energy/community safeguards, and rejection pending regional capacity review.

16. 49-cell welfare dictionary - compact validation version

This compact dictionary is extracted against the Canon Appendix AD role: it supports scoring interpretation and reviewer literacy. It does not modify gates, thresholds, equations, or claim boundaries.

Cell	Label	Plain meaning	Reviewer check
U1/D1	Personal Resources	An individual's economic security: income, savings, housing stability, and access to the material means of a decent life.	A reviewer checks whether the income/housing change is measured against a declared baseline cohort, not anecdote.
U1/D2	Personal Health	The individual's physical and mental health, safety, and bodily integrity.	A reviewer asks whether a claimed health gain rests on outcome data or only on inputs (e.g. clinics built $\neq$ health improved).
U1/D3	Relationships	The quality and stability of a person's close personal relationships and social connection.	A reviewer challenges whether a relationship claim is evidenced or inferred from a proxy like attendance.
U1/D4	Learning	The individual's access to education, skills, and the capacity to learn and develop.	A reviewer asks whether learning outcomes are demonstrated or only opportunity offered.
U1/D5	Personal Agency	The person's real ability to choose, refuse, consent, act, and shape their own life path.	A reviewer tests whether consent was genuine and revocable, or merely formal.
U1/D6	Purpose	The individual's sense of meaning, dignity, and worth in their life and work.	A reviewer distinguishes a genuine dignity effect from sentiment unsupported by experience data.
U1/D7	Local Conditions	The immediate physical environment a person lives in: air, noise, water, hazards, and surroundings.	A reviewer checks measured exposure change against the affected location, not a city-wide average.
U2/D1	Household Resources	The economic security of the household unit: shared income, assets, housing, and ability to meet needs.	A reviewer asks whether the household, not just the earner, is the measured unit.
U2/D2	Household Health	The collective physical and mental health and safety of household members, including dependents.	A reviewer checks that dependents and caregivers are scored separately, not blended into one household figure.
U2/D3	Family Cohesion	The strength, stability, and supportive quality of relationships within the household.	A reviewer asks whether a cohesion claim rests on outcomes or on assumed effects of a program.

Cell	Label	Plain meaning	Reviewer check
U2/D4	Shared Learning	The household's collective access to information, education, and learning capacity.	A reviewer checks access actually reaches the household, not just the area.
U2/D5	Household Agency	The household's collective ability to make decisions, plan, and control its own circumstances.	A reviewer tests whether household choice is real or constrained by conditions.
U2/D6	Family Meaning	The household's shared sense of identity, belonging, dignity, and purpose.	A reviewer separates a real meaning effect from program rhetoric.
U2/D7	Home Environment	The physical environmental quality of the home and its immediate setting.	A reviewer checks measured home conditions, not self-report alone.
U3/D1	Local Resources	The shared economic resources, infrastructure, and services available to a local community.	A reviewer asks whether the benefit is genuinely shared or captured by a subgroup.
U3/D2	Community Health	The collective physical and mental health and safety of the local population.	A reviewer checks whether subgroup harms are masked by a favourable community average.
U3/D3	Community Cohesion	The quality of trust, cooperation, belonging, safety, and relational fabric within a community.	A reviewer asks whether cohesion gains for some came via exclusion of others.
U3/D4	Local Knowledge	The community's shared knowledge, information access, and local expertise.	A reviewer checks consent and ownership of the knowledge claimed as preserved.
U3/D5	Community Agency	The community's collective capacity to organise, participate, and influence decisions affecting it.	A reviewer tests whether participation was substantive or a procedural formality.
U3/D6	Shared Meaning	The community's shared identity, culture, dignity, and sense of collective purpose.	A reviewer separates a genuine cultural effect from a symbolic gesture.
U3/D7	Local Ecology	The ecological condition of the community's local environment: green space, biodiversity, water, land.	A reviewer asks whether 'no local effect' was measured or merely assumed.

Cell	Label	Plain meaning	Reviewer check
U4/D1	Organizational Resources	An organization's operating means: finances, capital, staffing, and capacity to function and deliver.	A reviewer checks whether the organizational gain creates uncosted externalities in other cells (the classic 'operating resources up, U7 down' trap).
U4/D2	Organizational Safety	The safety of people within and affected by the organization: workers, users, and the public.	A reviewer checks whether reported safety reflects outcomes or only paperwork compliance.
U4/D3	Organizational Culture	The internal relational health of the organization: trust, fairness, inclusion, and cooperation.	A reviewer asks whether culture data is independent or self-reported by leadership.
U4/D4	Knowledge Systems	The organization's knowledge, data integrity, institutional memory, and learning systems.	A reviewer tests whether claimed integrity is verifiable or asserted.
U4/D5	Role Agency	The meaningful autonomy, voice, and fair treatment of individuals in their organizational roles.	A reviewer checks whether voice mechanisms produce outcomes or are decorative.
U4/D6	Mission Coherence	The alignment between the organization's stated purpose and its actual conduct and effects.	A reviewer asks whether coherence is evidenced by conduct or only by restated values.
U4/D7	Operational Footprint	The environmental impact of the organization's operations: emissions, waste, resource use, land.	A reviewer checks footprint against measured externalities, not offset claims alone.
U5/D1	Public Resources	The polity's fiscal and material capacity: public finances, infrastructure, and provisioning.	A reviewer checks whether gains are sustainable or borrowed from the future.
U5/D2	Population Health	The health, safety, and mortality outcomes of the whole population within a polity.	A reviewer checks whether worst-case scenarios were bounded with a tail measure, not averaged away.
U5/D3	Legitimacy	The rule of law, due process, accountability, and perceived legitimacy of public institutions.	A reviewer tests whether legitimacy is measured independently or claimed by the institution being assessed.

Cell	Label	Plain meaning	Reviewer check
U5/D4	Information Access	The polity's information environment: access, transparency, free expression, and epistemic integrity.	A reviewer checks whether transparency is real and usable or nominal.
U5/D5	Civil Agency	People's ability to participate in public life, exercise rights, contest authority, and influence governance.	A reviewer tests whether civic participation is enforceable or merely nominal, and whether any group is selectively excluded.
U5/D6	Public Meaning	Shared civic identity, dignity, social contract, and non-domination across the polity.	A reviewer separates symbolic recognition from changes people actually experience.
U5/D7	Public Environment	The environmental quality and ecological sustainability managed at the polity scale.	A reviewer checks whether environmental-justice distribution was assessed, not just regional totals.
U6/D1	Global Resources	Humanity's shared material base: global resource stocks, critical supply chains, and common infrastructure.	A reviewer checks whether 'global' claims rest on global data or extrapolated local figures.
U6/D2	Human Safety	The safety and survival of humanity at large, including existential and catastrophic risk.	A reviewer checks that tail risk was bounded with a worst-case measure and not diluted by expected-value framing.
U6/D3	Cooperation	The capacity of humanity to coordinate, cooperate, and maintain peaceful, stable relations at scale.	A reviewer asks whether cooperation is durable or a fragile short-term arrangement.
U6/D4	Civilizational Knowledge	Humanity's accumulated knowledge, science, and the integrity and safety of its knowledge systems.	A reviewer checks whether knowledge benefits were weighed against misuse and proliferation pathways.
U6/D5	Collective Agency	Humanity's capacity for self-determination and legitimate collective decision-making at the species scale.	A reviewer tests whether 'collective' agency includes the marginalised or only powerful actors.
U6/D6	Shared Purpose	Humanity's shared sense of meaning, moral direction, and long-term orientation toward flourishing.	A reviewer flags this cell as especially prone to rhetoric and demands concrete mechanisms.

Cell	Label	Plain meaning	Reviewer check
U6/D7	Planetary Conditions	The condition of planetary systems that support human civilisation and coordinated managing intelligence.	A reviewer checks whether irreversibility and worst-case bounds were modelled, not just expected outcomes.
U7/D1	Life-Support Resources	The biosphere's foundational resources that sustain life: soil, fresh water, clean air, fertility.	A reviewer checks whether renewal rates, not just current stocks, were assessed.
U7/D2	Ecosystem Health	The health, function, and viability of ecosystems and the species within them.	A reviewer checks whether ecosystem-function evidence exists or only a single charismatic indicator.
U7/D3	Biotic Relations	The integrity of relationships and interdependencies among living systems and between humans and nature.	A reviewer asks whether relational/cascade effects were modelled or ignored as 'no direct effect'.
U7/D4	Ecological Knowledge	Humanity's understanding and monitoring of ecological systems, including baseline and early-warning data.	A reviewer checks that absent ecological data is marked unknown, not scored as 0 ('no effect').
U7/D5	Resilience Capacity	The biosphere's capacity to absorb shocks, adapt, and recover from disturbance.	A reviewer asks whether proximity to thresholds was assessed, not just current condition.
U7/D6	Life Continuity	The continuity and persistence of life and biodiversity over time, including irreversibility of loss.	A reviewer checks whether irreversibility was treated as a tail constraint, not a discountable cost.
U7/D7	Ecological Integrity	The overall health, resilience, diversity, and continuity of ecosystems as living support systems.	A reviewer checks whether integrity is evidenced by ecosystem-level data or inferred from a single metric.

17. Recommended release ecosystem additions

Artifact	Role
RippleLogic_RLS_Validation_Protocol_v1.0	Study protocol, rater instructions, hypotheses, analysis plan, and revision rules.
RLS_Validation_Workbook_v0.1	Rater-entry and analysis-prep workbook.

Artifact	Role
RLS_Calibration_Note_v1.0	Future results document after the first scoring sprint.
Case_Packets/	Folder for frozen synthetic and real shadow-mode case packets.

18. Proposed README wording

Current status: RLS is release-ready as a structured residual welfare-ranking specification and audit method. Empirical validation of dimensional independence, inter-rater reliability, and decision-performance advantage is a next-stage research priority. The RLS Validation Protocol provides the study design for testing these claims through scored cases, independent raters, reliability statistics, and correlation/factor analysis of the 49-cell welfare field.

19. Completion definition for the first sprint

- At least 10 cases scored independently by at least 3 raters.
- All raw rater scores preserved.
- Completion, missingness, and disagreement statistics produced.
- IRR reported for score, ranking, and key classifications where applicable.
- Correlation or redundancy heatmap produced over the 49 cells.
- At least one scalar or reduced-matrix comparator run.
- Revision log produced with exact cells, anchors, or instructions needing repair.
- Public claim boundary updated: exploratory, preliminary, validated for limited domain, or not validated.